THE RATIO AND THE PRODUCT: TWO OBJECTS FROM THE EULER REFLECTION FORMULA AT $z = \frac{1}{4}$

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Abstract. The Euler reflection formula $\Gamma(\frac{1}{4})\,\Gamma(\frac{3}{4}) = \pi\sqrt{2}$ constrains but does not determine the pair $\{\Gamma(\frac{1}{4}),\, \Gamma(\frac{3}{4})\}$. Two independent objects encode the residual freedom: the *product* $\Gamma(\frac{1}{4})\,\Gamma(\frac{3}{4}) = \pi\sqrt{2}$, which is algebraic in π; and the *ratio* $G^{*} = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4}) = 2.958\,675\ldots$, which is algebraically independent of π. We show that $\log G^{*}$ absorbs every unsolved Dirichlet L-value at integer argument with rational coefficients, while π carries every solved one. The master quadratic $x^2 - 16G^{*2}\,x + 16G^{*3} = 0$ built from the ratio has roots $x_+ = 137.036$ and $x_- = 3.024$. Three centuries of mathematics studied the product. We argue that the ratio is where the coupling constants live, and that the two operations—product and ratio—correspond to the two modes of information processing that physics calls measurement and superposition.

1. Two Objects from One Formula

The Euler reflection formula at $z = \frac{1}{4}$ reads

$$\Gamma\big(\tfrac{1}{4}\big)\;\Gamma\big(\tfrac{3}{4}\big) \;=\; \frac{\pi}{\sin(\pi/4)} \;=\; \pi\sqrt{2}\,. \tag{1}$$

This constrains one combination of $\Gamma(\frac{1}{4})$ and $\Gamma(\frac{3}{4})$ but leaves one degree of freedom. That degree of freedom can be parameterized as:

$$\text{The product:}\quad \Gamma\big(\tfrac{1}{4}\big)\;\Gamma\big(\tfrac{3}{4}\big) \quad = \pi\sqrt{2}\,, \tag{2}$$

$$\text{The ratio:}\quad G^{*} \;:=\; \frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})} = 2.958\,675\,119\,188\,639\ldots \tag{3}$$

The product is fixed by (1). The ratio G^{*} is algebraically independent of π [1, 2]. Together they determine both Gamma values individually:

$$\Gamma\big(\tfrac{1}{4}\big) = \sqrt{G^{*}\,\pi\sqrt{2}}\,, \qquad \Gamma\big(\tfrac{3}{4}\big) = \sqrt{\frac{\pi\sqrt{2}}{G^{*}}}\,. \tag{4}$$

The product and the ratio are the two independent pieces of information in the pair $\{\Gamma(\frac{1}{4}),\, \Gamma(\frac{3}{4})\}$.

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2. What the Product Contains

The product $\pi\sqrt{2}$ lives in the world of solved constants. The anti-correlation theorem [3] establishes that at each integer $s \geq 2$, exactly one of $\zeta(s)$ and $\beta(s)$ reduces to a rational multiple of a power of π. The solved values are:

- Even ζ: $\zeta(2) = \pi^2/6$, $\zeta(4) = \pi^4/90$, ...
- Odd β: $\beta(1) = \pi/4$, $\beta(3) = \pi^3/32$, ...

These are the L-values that reduce to π. Physics built on π obtained Gaussian integration, statistical mechanics, the path integral measure—but not the coupling constants.

3. What the Ratio Contains

The ratio G^* absorbs the complementary set: the unsolved L-values.

Theorem 3.1 (Log G^* identity)**.**

$$\log G^* = \frac{\gamma + 3\log 2}{2} - \frac{G}{2} + \frac{7}{24}\zeta(3) - \frac{\beta(4)}{4} + \frac{31}{160}\zeta(5) - \cdots \tag{5}$$

where $G = \beta(2)$ is Catalan's constant, $\zeta(3)$ is Apéry's constant, and the general coefficients are: $\zeta(2m+1)$ has coefficient $(2^{2m+1}-1)/\big((2m+1)\cdot 2^{2m+1}\big)$; $\beta(2m)$ has coefficient $(-1)^{m+1}/(2m)$.

Every unsolved constant appears: Catalan, Apéry, $\beta(4)$, $\zeta(5)$, and the entire infinite tower. PSLQ analysis at 80+ digits finds no algebraic relation of degree ≤ 20 between any individual unsolved constant and $\{\pi, G^*, \varpi\}$.

The master quadratic built from G^* is

$$x^2 - 16G^{*2}\,x + 16G^{*3} = 0\,, \tag{6}$$

with coefficient $16 = |\mathrm{Aut}(E_i)|^2$ where $E_i\colon y^2 = x^3 - x$ is the CM elliptic curve with $j = 1728$. The roots are

$$x_+ = 137.036\,171\ldots\,, \qquad x_- = 3.023\,964\ldots \tag{7}$$

The ratio contains the fine structure constant. The product does not.

4. Product as Collapse, Ratio as Distinction

The two operations act differently on information:

Observation 4.1 (Information content). The product $\Gamma(\frac{1}{4})\cdot\Gamma(\frac{3}{4}) = \pi\sqrt{2}$ maps two algebraically independent transcendentals to a single value algebraic in π. The individual values are not recoverable from the product alone. The ratio $G^* = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$ preserves the relationship: given G^* and the product, both Gamma values are individually recoverable.

Conjecture 4.2 (Measurement–superposition correspondence)**.** *The product operation (information-destroying) corresponds to quantum measurement: $P = |\psi|^2 = \psi\cdot\psi^*$, which annihilates the complex phase. The ratio operation (information-preserving) corresponds to superposition: $\psi = a + bi$, which maintains the relationship between real and imaginary parts.*

Operation	*Information*	*Quantum mechanics*	*Interpretation*
Product (collapse)	*Destroyed*	$P = \|\psi\|^2$	*Measurement*
Ratio (distinction)	*Preserved*	$\psi = a + bi$	*Superposition*

The Born rule is a product: multiply ψ by ψ^* and the phase vanishes. Superposition is a ratio: the wave function maintains the relationship between its components. The phase—the thing proportional to i—is what distinguishes ψ from ψ^*.

5. The Imaginary Unit as Observer

The blind derivation chain [4] begins from a single axiom: i exists (i.e., $x^2 + 1 = 0$ has a solution). From this:

(1) $\mathbb{Z}[i]$ exists (Gaussian integers = square lattice).
(2) $E_i\colon y^2 = x^3 - x$ is the unique CM curve with $j = 1728$.
(3) $|\mathrm{Aut}(E_i)| = 4$, giving coefficient $16 = 4^2$.
(4) $G^* = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4})$ from the periods of E_i.
(5) The master quadratic (6) yields $x_+ = 137.036$.
(6) A one-loop lattice correction on $\mathbb{Z}[i]^3$ gives $x_+ = 137.036\,000$ (9.6 ppb from NIST).

The entire structure of the coupling constants follows from i.

Conjecture 5.1 (The observer as kernel)**.** *The projection* $\mathrm{Re}\colon \mathbb{C} \to \mathbb{R}$ *is the existence filter. Its kernel* $\ker(\mathrm{Re}) = i\mathbb{R}$ *is the space of quantities that participate in the dynamics but are projected out upon measurement. In quantum mechanics, this is the global phase of the wave function. We identify* $\ker(\mathrm{Re})$ *with the observer: the direction that is always perpendicular to every observable, always present in the evolution, and always annihilated when one asks for an outcome.*

6. Inversion: Physics as Constraint

If i is the ontological primitive—and the derivation chain suggests it is—then the relationship between mathematics and physics inverts:

> *Standard view:* Physics describes what exists. Mathematics is the language.
>
> *Proposed view:* i describes what exists (the full complex plane, including the observer). Physics describes what survives projection onto the real axis—what can be measured, observed, communicated.

The laws of physics, on this reading, are not prescriptions for what happens. They are constraints on what can be seen. The fine structure constant $\alpha^{-1} = 137.036$ is not a property of the electron; it is a property of the projection $\mathrm{Re}\colon \mathbb{C} \to \mathbb{R}$ applied to the arithmetic of $\mathbb{Z}[i]$.

Conjecture 6.1 (Alpha as projection cost)**.** $\alpha^{-1} = x_+$ *is the cost of collapsing the complex plane onto the real line at the scale set by the CM elliptic curve* E_i*.*

7. Why the Ratio Was Overlooked

Four historical factors explain the omission:

(1) **Period tradition.** Euler, Gauss, Legendre, and Jacobi studied the lemniscate and obtained $\varpi = \Gamma(\frac{1}{4})^2/(2\sqrt{2\pi})$. This involves the product, not the ratio.
(2) **π-centrism.** Writing constants in terms of π makes the product natural ($\pi\sqrt{2}$) and the ratio invisible (G^* has no π-based expression).
(3) **Solved-constant bias.** The product reduces to π (closed form). The ratio does not. Mathematics gravitates toward solvable quantities.
(4) **Trivial period ratio.** For E_i, the period ratio $\omega_2/\omega_1 = i$ is a known CM point; nobody extracts G^* from it because the ratio of *periods* is trivially i—the ratio of *Gamma values* is not.

8. Epistemic Status

Claim	Status	Basis
Reflection formula produces product and ratio	Theorem	Algebraic identity
G^* algebraically independent of π	Theorem	Nesterenko 1996
$\log G^*$ absorbs all unsolved L-values	Theorem	Identity (5)
Product = collapse, ratio = distinction	Conjecture	Structural analogy
Observer = $\ker(\mathrm{Re}) = i\mathbb{R}$	Conjecture	Interpretive
α^{-1} = projection cost	Conjecture	Interpretive

The mathematical content (Sections 1–3) is rigorous. The interpretive content (Sections 4–6) is conjecture. No experiment proposed here would distinguish this interpretation from standard quantum mechanics. The value is conceptual: it provides a structural reason *why* α has the value it does and *why* the observer is absent from standard physics.

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References

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